

Selection Sort

What are some strategies?

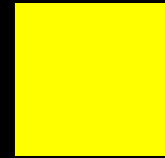
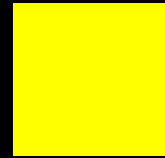
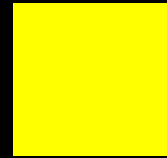
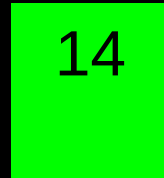
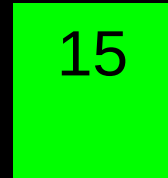
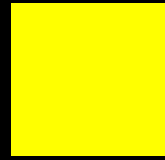
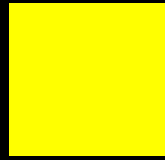
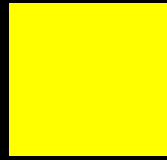
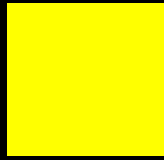
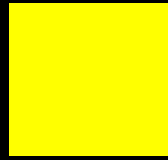
- Randomly choosing a card to face up (not systematic, and does not guarantee complete sorting)
- Systematically revealing the cards, one at a time (so which two to swap?)
- Others?

Unplugged Selection Sort

1-player game

- **Objective: Using Selection Sort, sort the cards (in ascending order) only revealing one card at any time.**
- **Equipment:**
 - 5 Number cards randomly arranged and faced down
 - 2 objects (pens will do), serving as pointer or demarcation
- **Rules:**
 - Player 1 (sorter)
 - Sorter can choose **only 1 card to be faced up at any point in time.**
 - Sorter can peek at the cards which are faced down, **one at a time.**
 - Cards can only be swapped
 - Once completed, get your tutor or classmate to check your method.

Demarcation
between sorted
and unsorted



Pointer marking
current smallest
value

Demarcation
between sorted
and unsorted



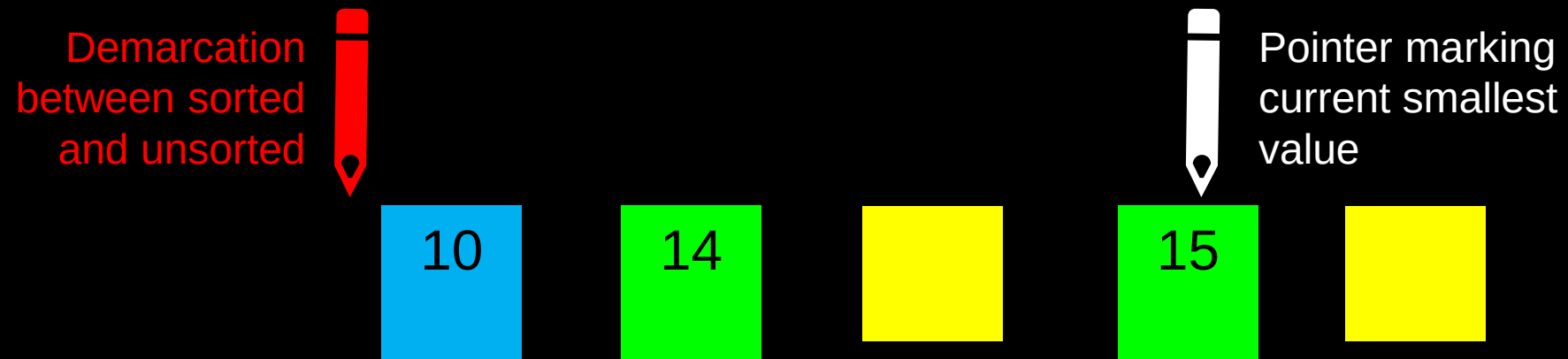
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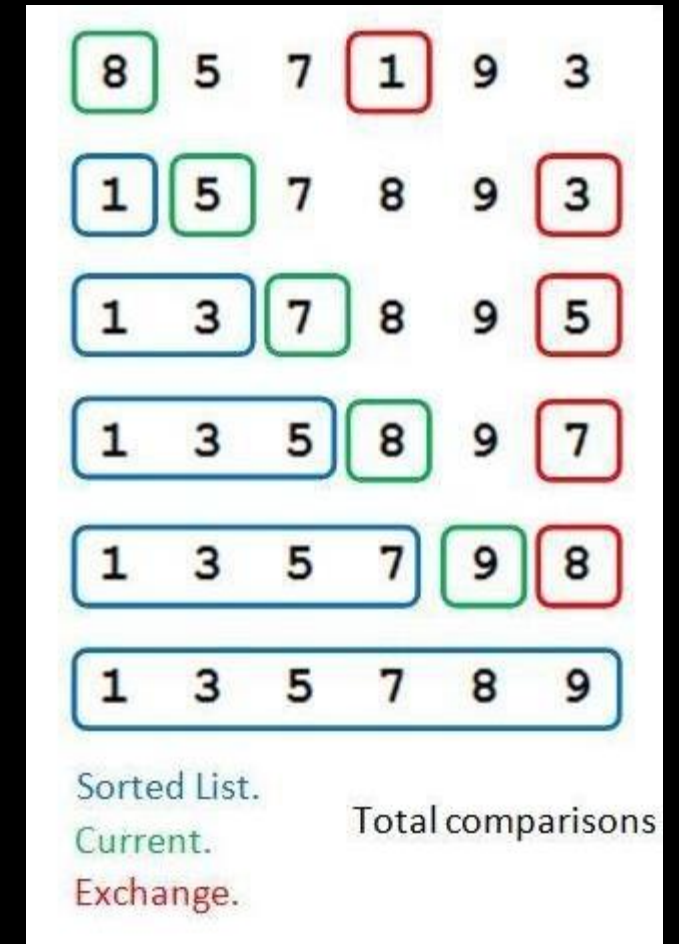
Pointer marking
current smallest
value

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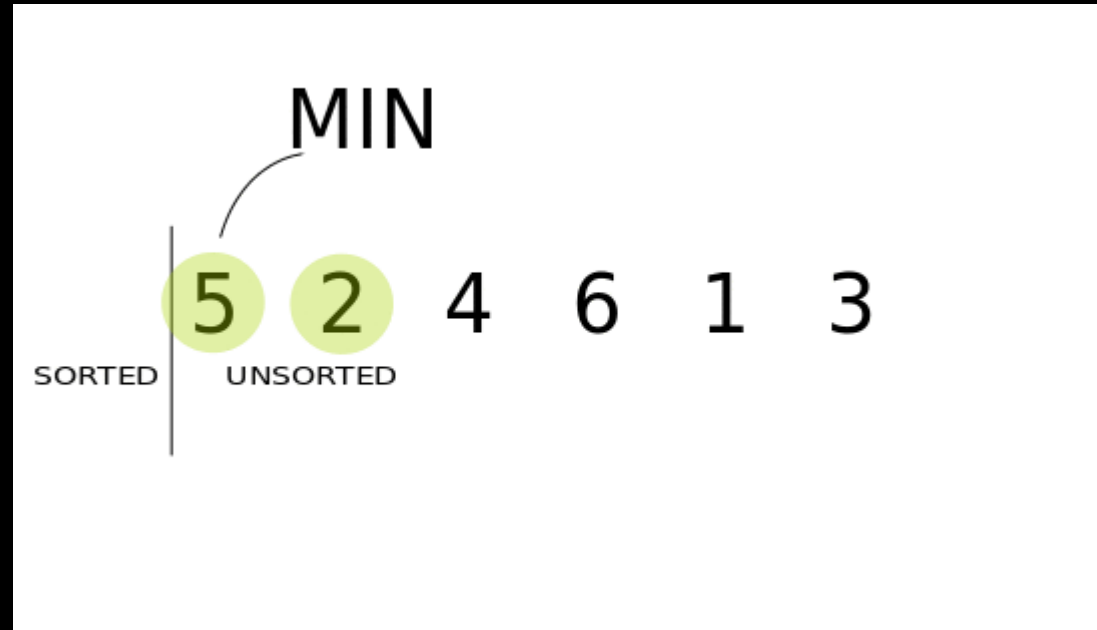


What is selection sort?

- Basic idea: Select the smallest element in the list and swap it with the leftmost unsorted element. That smallest element is considered as sorted.
- Step 1: Select the **smallest element** in the list and **swap** it with the **leftmost element**.
- Step 2: Select the **smallest element** in the **remaining** list and **swap** it with the **second element**
- **Repeating** this whole process of finding the **smallest element** in the **unsorted section** of the list and swapping it with the first element of the unsorted section until all the elements are sorted.



Selection sort Animation



<https://codepumpkin.com/selection-sort-algorithms/>

Is selection sort in-place?

- What does it mean to be in-place?
- Selection sort:
 - Elements doing mutual swap within the list
 - List is mutated
 - No additional memory space needed to store list during sorting
 - YES – Selection sort is in-place

Is selection sort stable?

- What does it mean to be stable?
- Consider a case where there are repeated elements within an unsorted list.
- [3, 1, 4, 5, 3, 6, 2]
- Stable:
- [1, 2, 3, 3, 4, 5, 6] #relative position remains unchanged
- Unstable:
- [1, 2, 3, 3, 4, 5, 6] #swap occurs for elements with same value
- Stability: Depends on its implementation, not the type of sort used

Is selection sort stable?

- Consider the following sequence: [4, 5, 3, 2, 4, 1]

Pass	Sorted index	Index 0	Index 1	Index 2	Index 3	Index 4	Index 5	Smallest value
1	0	4	5	3	2	4	1	
2								
3								
4								
5								
6								

4 and 4 have switched positions. Hence selection sort is unstable.

Technically, a stable selection sort is possible with some data structures like a singly linked list.

Order of growth **selection sort**?

- Best case: $O(n^2)$
 - Sum of AP: $(n-1) + (n-2) + (n-3) + \dots + 1$
- Average case: $O(n^2)$
- Worse case: $O(n^2)$
 - List is arranged in a reversed manner
 - Swapping occurs at every step